Fall 2026.

Physics 515. Statistical Mechanics.
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Homework # 2.

Due to October 8th, 2026.

You can use any source to solve the problems. However, your solutions should be presented in a very detailed and logical way. You need to understand every step of your solution and be able to explain it.

1. *Canonical ensemble*. Consider and ideal gas in the D-dimensional space if the energy ε and linear momentum $\vec{p}$ of the gas particles are connected by the following spectrum law, $\varepsilon = \alpha p^{\beta}$, where α, and β are constants. Prove that the gas equation of state is still $pV = NkT$. Hint: Transform an integral from the partition function into a dimensionless form rather than trying to evaluate it.

2. *Canonical ensemble*. Consider an ideal $3D$ gas of the volume V in the external field $U(\vec{r})$. Calculate the first non-zero correction to the gas heat capacity in the high-temperature limit $T \to \infty$. Hint: represent it as a function of average over the volume values of $U(\vec{r})$ and its powers.

3. *Canonical ensemble.* An ideal $3D$ gas is placed into a cylindrical vessel with the height H and external homogeneous gravitational field $U(\vec{r}) = mgz$. Find its Helmholtz free energy and heat capacity. Evaluate the low- and high-temperature limits of the results.

4. *Canonical ensemble*. An ideal $3D$ gas is placed into a central-symmetric trap with an impenetrable core of the radius R and external potential

$$U(\vec{r}) = \begin{cases} \infty, r \le R \\ U_0 \ln(r/R), \ r > R \end{cases} \cdot \ U_0 > 0$$

Find its Helmholtz free energy, heat capacity, and the maximal possible temperature.

5. *Canonical ensemble*. A quadruple moment of a molecule with the axial-symmetric electric charge distribution is given by the formula $Q_{ik} = Q(3n_i n_k - \delta_{ik})$, where n_i is a unit vector in the direction of the symmetry axis. In an external electric field with the electrostatic potential $\varphi = Q/r$ the energy of the molecule is equal to $U = Q_{ik}\sigma_{ik}$, where $\sigma_{ik} = \frac{1}{6}\frac{\partial^2 \varphi}{\partial x_i \partial x_k}$. Find an average quadruple momentum of the gas in the high-temperature limit (it is also the weak-field limit). Extra bonus: It could be useful to use the following formulae for average values: $\langle n_i n_k \rangle = (1/3)\delta_{ik}$, and $\langle n_i n_k n_m n_n \rangle = (1/15)(\delta_{ik}\delta_{mn} + \delta_{in}\delta_{km} + \delta_{im}\delta_{nk})$. Discuss the result.